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Appunti Tirocinio

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0.1 Problem Definition

0.1.1 Introduction

We consider an online variant of the Temporal Hitting Set problem under a sliding window model. To this end, we model the input as a temporal hypergraph $H = (V, E)$, where each hyperedge (S, t) consist of a subset of vertices $S \subseteq V$ and a discrete time step $t \in \mathbb{T}$. Given a time window of size ϵ , we define the set of active hyperedges at time t as:

$$W(t) = \{(S, t') \in E \mid t - \epsilon \leq t' \leq t\}.$$

At each time step t , the algorithm maintains a hitting set $X(t) \subseteq V$ such that:

$$\forall (S, t') \in W(t), \quad X(t) \cap S \neq \emptyset.$$

0.1.2 Objective

The goal is to maintain a feasible hitting set over time while optimizing the following criteria:

- **Solution size:** minimize $|X(t)|$ at each time step;
- **Update cost:** minimize the number of changes between consecutive solutions, measured as $\Delta X(t) = |X(t) \setminus X(t-1)| + |X(t-1) \setminus X(t)|$;
- **Computational efficiency:** minimize the time required to update the solution as the sliding window evolves.

Overall, the objective is to maintain a small and stable hitting set while efficiently handling the insertion and expiration of hyperedges induced by the sliding window.